(Please write your Enrollment Number) Enrollment No. 032

MID-TERM EXAMINATION

(CBCS)

B. Tech (IT/MAE/DMAM), 2nd Semester

(June, 2022)

Subject Code: BCS -110	Subject: Programming in C Language
Time : 1 ½ Hours	Maximum Marks : 30
Note: Q. 1 is compulsory. Attempt any two questions from the rest.	

Q1		(2.5*4)
	(a) Define ALU with an example.	
	(b) Convert 3GB into Bytes.	
	(c) Define the size of operator with an example.	
	(d) Define type casting with its associated example.	
Q2		(5,5)
	(a)Define Memory Hierarchy and explain its types with examples.	
	(b)Explain selection statements with syntax and flowchart using any one example with C- code.	
Q3		(5,5)
	(a) Define finite storage class with its types and example.	
	(b) Define C data types with their types and example.	
Q4		(5,5)
	(a)Explain Compiler, Interpreter, and Assembler with associated examples.	
	(b)Explain the concept of flowchart and algorithm with an associated algorithm to C – program using any suitable C -program as an example.	

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END TERM EXAMINATION

(CBCS)(SUBJECTIVE TYPE)

B. Tech IT/MAE/DMAM , 2nd Semester

(AUGUST, 2022)

Subject Code:BCS 110	Subject: Programming in C Language
Time :3 Hours	Maximum Marks :60
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.	

Q1		(2.5*8=20)
	(a) What is the difference between hardware and software?	
	(b) What is the use of registers in CPU?	
	(c) Differentiate between breakand continue statements.	
	(d) What is difference between while and do-while loops?	
	(e) Explain about various logical operators available in C language with examples	
	(f) What are preprocessor directives?	
	(g) Explain use getchar() and putchar() functions with example.	
	(h) What do you mean by array of structures?	
	UNIT-I	
Q2	(a) Differentiate between volatile and non-volatile memory? (b) Write any two differences between compilers and interpreters.	(5,5)
Q3	Write an algorithm and draw a flowchart to solve a quadratic equation.	(10)
	UNIT-II	
Q4	Writeaprogram tomake calculator using switch statement to perform addition, subtraction, multiplication and division operations.	(10)
Q5	(a) Write a program to find the sum of first and last digit of a number. (b) Write program to find maximum of two numbers using ternary operator.	(5,5)
	UNIT-III	
Q6	Write a program to find the largest number and the smallest number in a 3x3 matrix of integers.	(10)
Q7	Write a program to generate series of prime numbers between 1 to 50.	(10)
	UNIT-IV	
Q8	(a) Write a program to concatenate two strings without using strcat( ) function. (b) Difference between call by value and call by reference.	(5,5)
Q9	(a) Write a program to swap two numbers using pointers. (b) Describe how to declare and initialize structure and its members with an example.	(5,5)